

Name: AMRITANSU SINGH AMRITANSU SINGH | Reg. No: RA2511003011028

Unit “ 3 : Session “ 6 : SLO “ 2 | Stack Applications “ Tower of Hanoi

■ I. Fill in the Blanks

- TOWER OF HANOI is a classic problem that demonstrates the use of recursion with stacks.
- The minimum number of moves required to solve the Towers of Hanoi with n disks is $2^n - 1$.
- The recursive solution to Towers of Hanoi uses the STACK data structure to manage function calls.
- In the Towers of Hanoi, disks are moved from the source rod to the DESTINATION rod using an auxiliary rod.
- Each recursive call pushes a new STACK FRAME (activation record) onto the stack.
- The Towers of Hanoi involves TWO recursive steps (move $n-1$ disks twice, plus one move).
- The stack unwinds when the BASE case of the recursive function is reached.
- The base condition in Towers of Hanoi is when ONE ($n = 1$) disk is to be moved.
- Stacks in recursion help to TRACK (preserve) the order of operations in solving the puzzle.
- The auxiliary rod in the Towers of Hanoi acts as a TEMPORARY STORAGE (buffer) during disk transfers.